Vitaran Edutech

HINTS & SOLUTIONS

INTEGRATED PROGRAM

Biology

Class – 10

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1. Nutrition

Solutions

Worksheet – 1

1. **Correct Answer: C**

   Green plants are autotrophs because they use chlorophyll to capture sunlight and synthesize their own food through photosynthesis.

2. **Correct Answer: B**

   The hard palate is a bony structure formed primarily by the palatine processes of the maxilla and the horizontal plates of the palatine bones.

3. **Correct Answer: B**

   The frenulum linguae is the small fold of mucous membrane that secures the tongue to the floor of the mouth.

4. **Correct Answer: C**

   Incisors are the sharp, chisel-shaped front teeth specifically designed for cutting and biting off pieces of food.

5. **Correct Answer: C**

   Peyer's patches are organized lymphoid follicles found in the submucosa of the ileum, playing a key role in the immune system of the gut.

6. **Correct Answer: B**
   Premolars and molars are collectively called cheek teeth because of their position and function in grinding food against the cheeks.
7. **Correct Answer: B**
   The small intestine is anatomically divided into three parts: the C-shaped duodenum, the coiled jejunum, and the final section, the ileum.
8. **Correct Answer: B**
   The vermiform appendix is a narrow, finger-like vestigial tube that arises from the caecum at the beginning of the large intestine.
9. **Correct Answer: C**

   The large intestine is responsible for absorbing water and forming faeces, but it does not produce digestive enzymes; chemical digestion is largely finished by the time food reaches it.

10. **Correct Answer: B**

    The cardiac sphincter (also called the Lower Oesophageal Sphincter) acts as a valve between the Oesophagus and stomach to prevent acidic contents from moving upward.

Worksheet – 2

1. **Correct Answer: C**

   Lysozyme is an antibacterial enzyme that protects the oral cavity by chemically breaking down the cell walls of harmful bacteria

2. **Correct Answer: B**

   Hepatocytes are the main functional cells of the liver, performing the bulk of its metabolic, detoxifying, and secretory (bile) tasks.

3. **Correct Answer: C**

   Wharton's duct is the excretory duct that carries saliva from the submandibular gland to the floor of the mouth.

4. **Correct Answer: B**

   Bile salts act like detergents to break down large fat globules into tiny micelles, vastly increasing the surface area for lipase enzymes to work.

5. **Correct Answer: B**

   The pancreas is a heterocrine gland, secreting digestive enzymes into the intestine (exocrine) and hormones like insulin into the blood (endocrine).

6. **Correct Answer: B**
Approximately 30% of dietary starch is hydrolyzed into the disaccharide maltose by salivary amylase before reaching the stomach.
7. **Correct Answer: B**
Renin (also called chymosin) is a proteolytic enzyme found in gastric juice of infants that curdles milk proteins to aid their digestion.
8. **Correct Answer: B**
Parietal cells, located in the gastric glands, secrete hydrochloric acid (HCl) to activate enzymes and kill pathogens.
9. **Correct Answer: C**
Enterokinase is a brush-border enzyme in the small intestine that converts inactive trypsinogen into active trypsin, which then activates other proteases.
10. **Correct Answer: D**
The highly acidic environment created by Hydrochloric Acid (HCl) in the stomach triggers the conversion of inactive pepsinogen into the active protein-digesting enzyme, pepsin.

Worksheet-3

1. **Correct Answer: B**
Photosynthesis is the chemical process where plants convert CO_2 and H_2O into glucose using solar energy.
2. **Correct Answer: B**
Mesophyll cells contain the highest concentration of chloroplasts, making them the primary site for photosynthesis in a leaf.
3. **Correct Answer: D**
Similar to prokaryotes, chloroplast DNA is circular and "naked," meaning it is not associated with histone proteins.
4. **Correct Answer: C**
These two French pharmacists first isolated and named the green pigment "chlorophyll" in 1817.
5. **Correct Answer: A**
Carotenoids are accessory pigments consisting of orange-red carotenes and yellow xanthophylls.
6. **Correct Answer: C**
The light-dependent reactions convert solar energy into chemical energy in the form of ATP and NADPH to power the dark reaction.
7. **Correct Answer: B**
In this process, the photolysis (splitting) of water provides a continuous supply of electrons to replace those lost by Photosystem II.
8. **Correct Answer: D**
The law states that when a process depends on multiple factors, its rate is limited by the factor that is present in the least (minimum) amount.
9. **Correct Answer: A**
C_4 plants are more efficient at carbon fixation and reach photosynthetic saturation at much lower CO_2 levels than C_3 plants.
10. **Correct Answer: B**
While plants absorb massive amounts of water, more than 99% is lost through transpiration, leaving less than 1% for actual photosynthesis and growth.

Competitive Corner

SINGLE CORRECT OPTION TYPE

1. **Correct Answer: A**
Ptyalin is another name for salivary amylase, the enzyme in our mouth that begins the chemical breakdown of complex starch into simpler sugars.
2. **Correct Answer: B**
While light, temperature, and pH directly affect the chemical enzymes and energy of photosynthesis, humidity primarily affects transpiration rather than the photosynthetic rate itself.
3. **Correct Answer: D**
Maltose is a disaccharide (double sugar) specifically formed by the chemical bonding of two identical glucose molecules.
4. **Correct Answer: B**
ATP and NADPH are called "assimilatory powers" because they provide the chemical energy and reducing power needed to fix CO_2 into sugar during the dark reaction.

5. **Correct Answer: A**
   The hydrolysis of starch into maltose begins immediately in the mouth due to the action of salivary amylase (ptyalin).
6. **Correct Answer: D**
   Keeping a plant in the dark forces it to use up its stored starch for energy, ensuring that any starch found after the experiment was produced during the test.
7. **Correct Answer: A**
   Van Neil was the scientist who first proposed the general equation for photosynthesis, correctly identifying that the oxygen released comes from water, not carbon dioxide.
8. **Correct Answer: A**
   Trypsin is a proteolytic enzyme secreted by the pancreas that breaks down large protein molecules into smaller peptides and peptones in the small intestine.
9. **Correct Answer: D**
   Renin (found primarily in infants) is a specialized enzyme that curdles milk proteins (casein), allowing them to stay in the stomach longer for better digestion.
10. **Correct Answer: A**
    A balanced diet ensures the digestive system isn't overloaded and receives the necessary nutrients and fiber to function smoothly without distress.

ONE OR MORE THAN ONE CORRECT OPTION TYPE

11. **Correct Answer: B, C, D**

    The primary goal of the dark reaction (Calvin cycle) is to fix carbon dioxide into stable energy-rich glucose molecules with ADP and $NADP^+$

12. **Correct Answer: B, C**

    Chlorophyll pigments absorb light most efficiently in the blue and red wavelengths of the visible spectrum, while reflecting green light.

13. **Correct Answer: A, B, C**

    Vitamins A, D, E, and K are fat-soluble (stored in lipids), while Vitamin C and the B-complex are water-soluble.

14. **Correct Answer: A, B, D**
    The liver performs over 500 functions including cleaning toxins, storing energy as glycogen, and making bile; however, insulin is produced by the pancreas.
15. **Correct Answer: B, D**
    Sodium taurocholate and sodium glycocholate are the primary bile salts responsible for the emulsification of fats; bilirubin is a bile pigment.

ASSERTION AND REASON TYPE

16. **Correct Answer: C**
    While the small intestine wall does contain tubular glands (Crypts of Lieberkühn), DNase and RNase are actually enzymes secreted by the **pancreas**, not the intestinal glands.
17. **Correct Answer: D**
    The assertion is false because villi are absent in the large intestine (they are only needed in the small intestine for nutrient absorption); however, the reason is true as the large intestine does absorb water.
18. **Correct Answer: A**
    The muscular layers of the alimentary canal perform peristalsis and churning movements, which are essential for physically mixing food with digestive juices and enzymes.
19. **Correct Answer: B**
    It is true that the light reaction produces ATP/NADPH and the dark reaction uses them, but the *usage* in the dark reaction is not the *reason why* they are produced in the first place.

MATRIX MATCHING TYPE

20. **Correct Answer: C**
    a-r, b-p, c-s, d-q
21. **Correct Answer: A**
    a–r, b–p, c–s, d–t, e–q

COMPREHENSION TYPE

Passage – I

22. **Correct answer**
    1. **C -** Chloroplasts contain the green pigment chlorophyll and the necessary

enzymes to convert solar energy into chemical energy during photosynthesis.

2. **B -** Chlorophyll is the primary pigment for absorbing light, while carotenoids act as accessory pigments that broaden the spectrum of light used and protect the plant from solar damage.
3. **C -** The dark reaction (or Calvin Cycle) uses the chemical energy (ATP and NADPH) produced during the light reaction to fix carbon dioxide into stable organic molecules like glucose.

Passage – II

23. Correct answer

1. **C -** Digestion begins in the mouth (buccal cavity) where teeth perform mechanical breakdown and saliva initiates the chemical breakdown of carbohydrates.
2. **C -** Salivary amylase (also known as ptyalin) is secreted by the salivary glands and begins the process of converting complex starch into simpler sugars like maltose.
3. **C -** The Oesophagus (food pipe) uses rhythmic muscular contractions called peristalsis to push the chewed food (bulus) from the throat down into the stomach.

Previous Year Questions

1. Correct Answer: B

In humans, the milk (deciduous) dentition includes 8 incisors, 4 canines, and 8 molars; pre-molars are only present in the permanent set.

2. Correct Answer: C

Pancreatic juice contains inactive enzymes like trypsinogen and procarboxypeptidase, along with amylase for starch and lipase for fats.

3. Correct Answer: C

Rennin is a proteolytic enzyme found in the gastric juice of infants that initiates milk digestion by curdling milk proteins.

4. Correct Answer: D

The ileum is the longest and most highly coiled portion of the small intestine, specialized for nutrient absorption.

5. Correct Answer: D

Statement (ii) is false because assimilatory power is produced in the light phase, and (iv) is false because radiant energy is converted to chemical energy, not vice versa.

6. Correct Answer: C

Rennin is specifically secreted by the gastric glands in the stomach to aid in the digestion of milk proteins in infants.

7. Correct Answer: C

Electrons move from Photosystem II to the cytochrome b_6f complex, then to Photosystem I, and finally reduce $NADP^+$ to NADPH.

8. Correct Answer: C

The water-splitting complex is associated with the inner side of the thylakoid membrane, so hydrogen ions are released into the lumen.

9. Correct Answer: C

Cyclic photophosphorylation involves only Photosystem I and produces only ATP, without the evolution of oxygen or the production of NADPH.

10. Correct Answer: C

As proven by Van Niel, the oxygen released during photosynthesis originates from the photolysis (splitting) of water molecules.

2. Respiration

Solutions

Worksheet – 1

1. **Correct Answer: C Catabolic.**
   Respiration is a catabolic process because it involves the breakdown of complex molecules (like glucose) into simpler substances, releasing energy in the process.
2. **Correct Answer: B Absence of oxygen.**
   Anaerobic respiration takes place when oxygen is not available, allowing organisms or cells to release energy without using oxygen.
3. **Correct Answer: A Trachea**
   The epiglottis acts like a flap that covers the opening of the trachea (windpipe) during swallowing, preventing food from entering the respiratory tract.
4. **Correct Answer: B Carbon dioxide, water, and ATP**
   Cellular respiration breaks down glucose in the presence (or absence) of oxygen to release energy in the form of ATP, producing carbon dioxide and water as by-products.
5. **Correct Answer: C Vestibular part**
   The vestibular part of the nasal chamber contains hair and mucus, which help trap dust and foreign particles entering with the air.
6. **Correct Answer: C Smell**
   The olfactory epithelium contains sensory receptors that detect odours, making it responsible for the sense of smell.
7. **Correct Answer: C Internal nostrils**
   The nasal chambers open into the nasopharynx through the internal nostrils, also called choanae.
8. **Correct Answer: C Windpipe**
   The trachea is commonly known as the windpipe because it carries air to and from the lungs.
9. **Correct Answer: B Pulmonary tree**
   The pulmonary tree refers to the extensive branching of the trachea into bronchi, bronchioles, and alveoli, resembling an inverted tree inside the lungs.
10. **The Correct Answer: C 21% oxygen**
    Atmospheric air normally contains about 21% oxygen, which is essential for aerobic respiration.

Worksheet-2

1. **The Correct Answer: B Spirometer**

   A spirometer is used to measure the volume of air inhaled and exhaled during breathing, helping assess lung function.

2. **The Correct Answer: B Inspiration and expiration**

   Breathing consists of two phases:

   Inspiration (taking air into the lungs)

   Expiration (expelling air out of the lungs)

3. **The Correct Answer: C Less than atmospheric pressure**

   During inspiration, the intra-pulmonary pressure decreases below atmospheric pressure, causing air to flow into the lungs.

4. **The Correct Answer: C Inspiratory Capacity**

   (IC) is the maximum volume of air that can be inhaled after a normal expiration, and it is the sum of Tidal Volume (TV) and Inspiratory Reserve Volume (IRV).

5. **The Correct Answer: D TV + IRV + ERV**

   Vital Capacity (VC) is the maximum volume of air that can be exhaled after a deep inspiration, and it includes Tidal Volume (TV), Inspiratory Reserve Volume (IRV), and Expiratory Reserve Volume (ERV).

6. **The Correct Answer: C Expiration**
During expiration, the diaphragm relaxes and returns to its dome-shaped position, and the sternum moves back to normal, reducing thoracic volume and pushing air out of the lungs.
7. **The Correct Answer: C Tidal volume**
Tidal volume is the amount of air inhaled or exhaled during normal, quiet breathing.
8. **Correct Answer: D Forced inspiration**
Total lung capacity (TLC) is the maximum volume of air the lungs can hold after a person takes the deepest breath possible.
9. **Correct Answer: C Increases above atmospheric pressure**
Air flows from areas of high pressure to low pressure. During **expiration**, the thoracic cavity decreases in volume, which causes the **intra-pulmonary pressure** to rise above the pressure of the outside air, forcing it out of the lungs.
10. **Correct Answer: B Lungs and surrounding atmosphere**
Breathing is driven by a pressure gradient between the air inside the **lungs** (intra-pulmonary pressure) and the **atmospheric air** outside the body.

Worksheet -3

1. **Correct Answer: D 3%**
Only a small fraction of oxygen, approximately **3%**, is transported in a dissolved state within the blood plasma.
2. **Correct Answer: C Red blood cells**
About **97%** of oxygen is transported by Red blood cells (RBCs) because they contain **haemoglobin**, which has a high affinity for oxygen.
3. **Correct Answer: C Carbon dioxide**
In plants, **respiration** involves breaking down glucose to release energy, producing **carbon dioxide** and water as by-products.
4. **Correct Answer: C During day and night**
Respiration is a continuous process required for cellular energy and occurs **all the time** (both day and night) in plants.
5. **Correct Answer: C Oxyhaemoglobin**
Oxygen binds reversibly with haemoglobin in the lungs to form a bright red compound called oxyhaemoglobin.
6. **Correct Answer: B Low oxygen partial pressure**
At the tissue level, the **partial pressure of oxygen** is low because cells are using it up. This low pressure triggers **oxyhaemoglobin** to dissociate and release oxygen.
7. **Correct Answer: C Stomata and lenticels**
Gaseous exchange in plants occurs through tiny pores called stomata (mostly on leaves) and lenticels (on the woody stems of older plants).
8. **Correct Answer: B RBCs**
Most **bicarbonate ions** are formed inside **Red blood cells** because they contain the enzyme **carbonic anhydrase**, which rapidly converts and water into carbonic acid, which then dissociates into bicarbonate.
9. **Correct Answer: B Oxidation of carbohydrates**
Oxygen is primarily used in plants to **oxidise carbohydrates** (glucose) to release the chemical energy (ATP) needed for growth and maintenance.
10. **Correct Answer: C Three**
Carbon dioxide is transported in the blood in **three** distinct forms: dissolved in plasma (7%), as bicarbonate ions (70%), and as carbamino-haemoglobin (20-25%)

Competitive Corner

SINGLE CORRECT OPTION TYPE

1. **Correct Answer: C Final stage**
Respiration is the final stage in food utilization where digested nutrients are oxidized within cells to release energy (ATP).
2. **Correct Answer: B Nostrils**
The nostrils (external nares) are the two external openings that allow air to enter the nasal cavity.

3. **Correct Answer: C Iron-containing red pigment**
   Haemoglobin is a complex protein found in RBCs that contains **iron** which gives blood its red color and allows it to bind to oxygen.
4. **Correct Answer: C Moderate respiratory rhythm**
   The pneumotaxic centre, located in the pons, primarily functions to limit inspiration, thereby regulating the respiratory rate and pattern.
5. **Correct Answer: C Contraction of diaphragm and intercostal muscles**
   Inspiration is "active" because it requires ATP for the contraction of the diaphragm and external intercostal muscles to expand the thoracic cavity.
6. **Correct Answer: B Maximising gaseous exchange**
   The primary purpose of **breathing** (ventilation) is to constantly refresh the air in the lungs to maintain the concentration gradients needed for efficient gas exchange.
7. **Correct Answer: C CO_2 is immediately used in photosynthesis**
   In the daytime, plants perform both respiration and photosynthesis. The produced by respiration is recycled for **photosynthesis**, making the net gas exchange less obvious.
8. **Correct Answer: C Medulla oblongata**
   The primary **respiratory rhythm centre**, which is responsible for the basic rhythm of breathing, is located in the medulla oblongata.
9. **Correct Answer: C Reversible association**
   Oxygen binds to haemoglobin in a reversible manner; it binds where oxygen pressure is high (lungs) and detaches where it is low (tissues).
10. **Correct Answer: C Breathing rhythm**
    Because the medulla oblongata houses the **respiratory rhythm centre**, damage to this area would immediately disrupt or stop the automatic breathing process.

ONE OR MORE THAN ONE CORRECT OPTION TYPE

11. **Correct Answer: B) Absence of oxygen, C) Lower energy yield, and D) Occurrence in yeast and bacteria**

    Anaerobic respiration is a type of cellular respiration that occurs without oxygen. It is characterized by:

    1. Absence of oxygen: It takes place in environments where molecular oxygen is unavailable.
    2. Lower energy yield: Because glucose is only partially broken down, it produces much less ATP (energy) compared to aerobic respiration.
    3. Occurrence in specific organisms: It is common in microorganisms like yeast and certain bacteria.

12. **Correct Answer: A) Pulmonary ventilation, B) Transport of gases by blood, and C) Diffusion of gases at tissues**

    The process of respiration in humans involves several integrated steps:

    1. **Pulmonary ventilation**: Breathing air in and out of the lungs.
    2. **Transport of gases**: Moving oxygen and carbon dioxide through the circulatory system.
    3. **Diffusion at tissues**: The exchange of and between the blood and body cells.

13. **Correct Answer: A) Oxygen and B) Carbon dioxide**

    Blood serves as the primary transport medium for respiratory gases:

    1. **Oxygen**: Carried mostly by haemoglobin in red blood cells.
    2. **Carbon dioxide**: Carried as bicarbonate ions, bound to haemoglobin, or dissolved in plasma.

14. **Correct Answer: A) Formation of carbonic acid, B) Conversion of carbonic acid to bicarbonate, and D) Rapid CO_2 transport**

The enzyme **carbonic anhydrase** is essential for carbon dioxide transport:

1. It catalyzes the formation of carbonic acid from and water.
2. It facilitates the rapid conversion of this acid into **bicarbonate ions** and hydrogen ions This allows for the efficient and rapid transport of from tissues to the lungs.

15. **Correct Answer: A) Are actively growing, B) Require more energy, and D) Contain meristematic tissue**

In plants, the rate of respiration is highest in regions with high metabolic activity:

1. **Actively growing parts**: Developing buds, germinating seeds, and young leaves need more energy for synthesis.
2. **Meristematic tissue**: These areas (like root and shoot tips) undergo constant cell division, which is energy-intensive.

ASSERTION AND REASON TYPE

16. **Correct Answer: D**

Breathing is just the physical act of swapping gases, while respiration is the chemical process of making energy.

17. **Correct Answer: A**

the respiratory system looks like a tree because the airways branch into smaller and smaller tubes like twigs.

18. **Correct Answer: A**

Less energy is made because food is only partially broken down when oxygen is not available to help.

19. **Correct Answer: A**

Pleural fluid acts like a lubricant to stop the lungs from rubbing painfully against the chest wall.

MATRIX MATCHING TYPE

20. **Correct Answer: A**
21. **Correct Answer: A**

COMPREHENSION TYPE

PASSAGE-1

22. **Correct Answer: C**
Relaxation of respiratory muscles
23. **Correct Answer: C**
Increases allowing air to enter
24. **Correct Answer: D**
Increase in intrapulmonary pressure

PASSAGE-2

25. **Correct Answer: C**
Energy released during respiration
26. **Correct Answer: C**
Energy released during respiration
27. **Correct Answer: C**
Stomata and lenticels

PREVIOUS YEAR QUESTIONS

1. **Correct Answer: B 5 ml**

Under normal physiological conditions, every 100 ml of oxygenated blood carries about 20 ml of , but only delivers approximately 5 ml of it to the tissues.

2. **Correct Answer: B residual volume**

Lungs do not collapse even after a forceful expiration due to **Residual Volume (RV)**, which is the air that always remains in the lungs (approx. 1100–1200 mL) to keep alveoli inflated.

3. **Correct Answer: A**
Rising CO2 concentration

The urge to breathe when holding one's breath is primarily triggered by **Rising concentration** in the blood, which lowers pH and stimulates the respiratory centre in the brain.

4. **Correct Answer: D) Higher H+ conc. in alveoli favours the formation of oxyhaemoglobin**

5. **Correct Answer: (b) A – III; B – I; C – IV; D – II**

3. Transportation

Solutions

Worksheet – 1

1. **Correct Answer: B.**
   Circulatory system carry nutrient, oxygen to the body parts and take waste as carbon dioxide from different part to the appropriate excretory organ
2. **Correct Answer: C**
   surface antigen present on RBCs decide whether it's A, B or O blood group
3. **Correct Answer: C**
   Plasma makes up about 55% of total blood volume.
4. **Correct Answer: A**
   Blood group B has B antigens on RBCs and anti-A antibodies in plasma.
5. **Correct Answer: B**
   Serum is plasma after removal of clotting factors.
6. **Correct Answer: B**
   Erythropoiesis specifically refers to RBC formation.
7. **Correct Answer: C**
   Thrombin converts soluble fibrinogen into insoluble fibrin.
8. **Correct Answer: C**
   Fibrin is formed from fibrinogen during blood clotting.
9. **Correct Answer: C**
   Low platelet count impairs clotting, causing bleeding.
10. **Correct Answer: D**
    Lymphocytes are the main immune cells present in lymph.

Worksheet – 2

1. **Correct Answer: C**

   In a closed system, blood always flows within blood vessels.

2. **Correct Answer: D**

   here is no phase called pulmonary systole in the cardiac cycle.

3. **Correct Answer: C**

   The sinoatrial node initiates atrial systole as the natural pacemaker

4. **Correct Answer: C**

   crocodile is the exception in reptiles with fully four chambered Heart.

5. **Correct Answer: B**

   Semilunar valves open when ventricular pressure exceeds arterial pressure.

6. **Correct Answer: C**

   The average stroke volume in humans is about 70 ml per heartbeat.

7. **Correct Answer: C**

   The pericardium is outermost double-walled protective sac around the heart.

8. **Correct Answer: B**

   The cardiac center in the medulla regulates heart rate.

9. **Correct Answer: B**

   The inter-atrial septum separates the right and left atria.

10. **Correct Answer: C**

    The left ventricle has thicker walls to pump blood at high pressure to the body.

Worksheet-3

1. **Correct Answer: A**

   Phloem transports food along with hormones and water to different parts of the plant.

2. **Correct Answer: B**

   Calcium is largely immobile once deposited in plant tissues.

3. **Correct Answer: B**

   Transpiration pull creates a suction force that lifts water upward in plants.

4. **Correct Answer: C**

   Phloem transports food from source to sink in both upward and downward directions.

5. **Correct Answer: B**

   Xylem is the main tissue responsible for long-distance water transport.

6. **Correct Answer: C**

   Active transport uses energy (ATP) to move substances against the gradient.

7. **Correct Answer: B**

   Diffusion is a passive process and does not require energy.

8. **Correct Answer: C**

   Guttation is the exudation of liquid water droplets from leaf margins.

9. **Correct Answer: C**

   Source and sink roles change depending on the plant's growth stage.

10. **Correct Answer: C**

    Water is mainly absorbed by root hairs through osmosis.

Competitive Corner

SINGLE CORRECT OPTION TYPE

1. **Correct Answer: A**

   Left atrium → left ventricle → aorta – Oxygenated blood flows from the left atrium to the left ventricle and is pumped to the body through the aorta.

2. **Correct Answer: B**

   Agranulocytes lack cytoplasmic granules and include lymphocytes and monocytes.

3. **Correct Answer: B**

   As turgidity rises, the cell wall exerts greater wall pressure.

4. **Correct Answer: A**

   Water enters the cell from a less concentrated solution by endosmosis.

5. **Correct Answer: D**

   Calcium ions are essential for several steps in blood clotting.

6. **Correct Answer: A**

   Cardiac output is the volume of blood pumped by the heart per minute.

7. **Correct Answer: D**

   The pulmonary vein carries oxygenated blood from the lungs to the left atrium.

8. **Correct Answer: D**

   The upward movement of water and minerals in plants is called ascent of sap.

9. **Correct Answer: A, B, C**

   Thymus, Tonsils, Lymphatic nodes - All these are lymphoid tissues involved in immunity.

10. **Correct Answer: C**

    Only statement iv is correct; the others incorrectly describe heart chambers and partitions.

ONE OR MORE THAN ONE CORRECT OPTION TYPE

11. **Correct Answer: A, B, C**
    all Calcium ions, Platelet factor and vitamin K are essential for blood coagulation; calcium activates clotting factors and platelet factor helps in clot formation (heparin prevents clotting, vitamin K acts indirectly).

12. **Correct Answer: D**
    Guard cells change their shape by absorbing or losing water (turgidity), which physically opens or closes the stomatal pore to regulate gas exchange and water loss.

13. **Correct Answer: ii**
    In an open system, blood bathes organs directly in open cavities (sinuses); because there are no closed vessels to "steer" the flow, the body cannot precisely regulate blood supply to specific organs.

14. **Correct Answer: A**
    While all factors matter, humidity is the most direct regulator because transpiration depends on the diffusion gradient; if the outside air is already saturated with water, evaporation from the leaf slows down significantly.

15. **Correct Answer: i and iv**

Statement **i** is true as "thrombocytes" is the scientific name for platelets; statement **iv** is true because a low platelet count prevents the formation of a "plug," leading to uncontrolled bleeding.

ASSERTION AND REASON TYPE

16. **Correct Answer: A**

Diapedesis is the specific process where WBCs squeeze through the capillary walls (endothelium) to reach and fight infections at the wound site.

17. **Correct Answer: A**

Serum is defined as plasma minus clotting factors (like fibrinogen); therefore, it physically lacks the proteins necessary to form a clot.

18. **Correct Answer: C**

Serum is the straw-colored fluid left after a clot forms, and Fibrin is the "net" created from soluble fibrinogen that traps blood cells to form that clot.

19. **Correct Answer: A**

For the atria to contract and pump blood effectively into the ventricles (assertion), they must first receive the electrical signal (Action Potential) generated by the SA node (reason).

20. **Correct Answer: C**

Phloem transport is bi-directional (moving sugar to roots or fruits), but the reason is false because it describes xylem, which moves water strictly in a unidirectional, upward flow.

MATRIX MATCHING TYPE

21. **Correct Answer: D**

A – iii, B – iv, C – i, D – ii

22. **Correct Answer: B**

1 – r, 2 – p, 3 – s, 4 – q

23. **Correct Answer: B**

A – ii, B – iii, C – iv, D – i

COMPREHENSION TYPE

24. **Correct Answer:**
    1. B - systole refers to contraction; therefore, atrial systole is the specific phase where the top chambers contract to top off the blood in the ventricles.
    2. B - "Lubb" sound (first heart sound) is caused by the forceful closure of the Atrioventricular (mitral and tricuspid) valves at the start of ventricular contraction to prevent backflow.
    3. B - Stroke volume is the specific amount of blood (approximately 70 ml) ejected by a single ventricle during one contraction of the heart.

25. **Correct Answer:**
    1. C - Erythrocytes (Red Blood Cells) are packed with Haemoglobin, the iron-containing protein specialized for transporting oxygen throughout the body.
    2. D - Platelets (Thrombocytes) are the cell fragments that aggregate at a wound site and release chemicals to initiate the blood clotting process.
    3. D - White Blood Cells (Leucocytes) are the primary soldiers of the immune system, responsible for identifying and destroying foreign pathogens like bacteria and viruses.

Previous Year Questions

1. **Correct Answer: D**
This statement is incorrect because the paragraph actually switches their roles; blood from the right ventricle flows to the lungs via the *pulmonary artery*, and from the left ventricle via the *Aorta*.

2. **Correct Answer: D**
While primarily for water, xylem also carries dissolved minerals, organic nitrogen compounds (like amino acids), and plant hormones.

3. **Correct Answer: B**
   The heartbeat starts at the SA-node (pacemaker), moves to the AV-node, travels down the Bundle of His, and spreads through Purkinje fibres to contract the ventricles.
4. **Correct Answer: A**
   Pistia is a free-floating aquatic plant that absorbs water directly through its leaf surfaces, making its roots largely insignificant for water intake.
5. **Correct Answer: C**
   At night, when transpiration is low, positive root pressure pushes liquid water out through specialized pores called hydathodes (guttation).
6. **Correct Answer: C**
   A decrease in pH (Bohr effect) shifts the oxygen dissociation curve to the right, causing hemoglobin to release oxygen more easily to tissues.
7. **Correct Answer: C**
   Blood is a connective tissue composed of a liquid matrix (plasma) and formed elements (RBCs, WBCs, and platelets).
8. **Correct Answer: B**
   By removing the bark (including phloem), researchers observed that food transport was blocked, while water transport (xylem) remained intact.
9. **Correct Answer: A**
   P-wave represents atrial depolarization (iv),
   QRS is ventricular depolarization (i),
   T-wave is ventricular repolarization (ii),
   reduced T-wave indicates ischemia (iii).
10. **Correct Answer: C**
    In adults, the production of red blood cells (erythropoiesis) occurs primarily in the red bone marrow of long and flat bones.

4. Excretion

Solutions

Worksheet – 1

1. **Correct Answer: B**
   Excretion specifically refers to the removal of toxic by-products (like urea or CO_2) generated by chemical reactions within cells.
2. **Correct Answer: C**
   Deamination is the process where the amino group is removed from amino acids in the liver, resulting in the production of ammonia.
3. **Correct Answer: C**
   Ammonotelic animals, such as many aquatic invertebrates and bony fishes, excrete nitrogenous waste primarily as highly soluble ammonia.
4. **Correct Answer: C**
   Small amounts of urea are retained in the kidney's interstitial fluid to maintain high osmotic concentration, which helps in water reabsorption.
5. **Correct Answer: C**
   Uric acid is the least toxic and least soluble nitrogenous waste, excreted as a thick paste or pellet to maximize water conservation.
6. **Correct Answer: C**
   These specialized organs (like kidneys, lungs, and skin) are dedicated to filtering and removing metabolic wastes from the body.
7. **Correct Answer: B**
   At night, photosynthesis stops while respiration continues, making carbon dioxide the primary gas released by the plant.
8. **Correct Answer: C**
   Transpiration is the evaporation of water through the stomata of leaves, which helps regulate temperature and move minerals.
9. **Correct Answer: C**
   These trees store metabolic wastes in their leaves and successfully eliminate them when the leaves are shed during autumn.
10. **Correct Answer: A**
    Sponges (Porifera) lack specialized organs and rely on simple diffusion across their cell surfaces to remove wastes into the surrounding water.

Worksheet – 2

1. **Correct Answer: B**
   These four components work together to filter blood, transport urine, store it, and finally expel it from the body.
2. **Correct Answer: C**
   The large size and position of the liver on the right side of the abdominal cavity push the right kidney slightly lower than the left.
3. **Correct Answer: C**
   Each human kidney contains approximately 1 to 1.2 million microscopic filtering units known as nephrons.
4. **Correct Answer: B**
   The nephron is the smallest unit capable of performing all the kidney's functions, including filtration, reabsorption, and secretion.
5. **Correct Answer: B**
   The outer layer is the renal cortex (pale in color), and the inner layer is the renal medulla (containing the pyramids).
6. **Correct Answer: C**
   These are triangular, cone-shaped tissue masses in the medulla that contain the loops of Henle and collecting ducts.
7. **Correct Answer: D**
   This specialized tissue (also called urothelium) is designed to stretch and expand as urine passes through or is stored.

8. **Correct Answer: C**
The urethra is the muscular tube that carries urine from the bladder to the outside of the body during urination.
9. **Correct Answer: C**
Filtrate moves from Bowman's capsule to the Proximal Convoluted Tubule, through the Loop of Henle, to the Distal Convoluted Tubule, and finally into a collecting duct.
10. **Correct Answer: B**
While the Loop of Henle and collecting ducts dip into the medulla, the main "body" of the nephron (corpuscle and convoluted tubules) stays in the cortex.

Worksheet-3

1. **Correct Answer: C**
The first step of urine formation is the high-pressure filtration of blood through the glomerular capillaries into the Bowman's capsule.
2. **Correct Answer: D**
High threshold substances are those that are completely reabsorbed from the filtrate under normal physiological conditions because they are vital to the body.
3. **Correct Answer: D**
Vasopressin (also known as ADH) increases water reabsorption in the collecting ducts, which results in the production of concentrated (hypertonic) urine.
4. **Correct Answer: C**
The final volume of urine is determined by the amount filtered, minus what the body takes back, plus any additional wastes added by the tubules.
5. **Correct Answer: B**
As secondary excretory organs, the lungs eliminate gaseous metabolic wastes like CO_2 and moisture through the process of exhalation.
6. **Correct Answer: C**
In clinical practice, blood is usually pumped out from a convenient artery like the radial artery in the wrist to be cleaned by the dialyzer.
7. **Correct Answer: B**
This is the initial liquid that passes through the filter; it contains water, glucose, salts, and urea but lacks blood cells and large proteins.
8. **Correct Answer: A**
ADH (Antidiuretic Hormone) prevents excessive water loss by signaling the kidneys to conserve water, leading to more concentrated urine.
9. **Correct Answer: C**
The cellophane tubes in an artificial kidney act as a semi-permeable membrane, allowing small waste molecules to pass through while keeping blood cells inside.
10. **Correct Answer: B**
Haemodialysis relies on a concentration gradient to move nitrogenous wastes from the blood into the dialyzing fluid through a microscopic filter.

Competitive Corner

SINGLE CORRECT OPTION TYPE

1. **Correct Answer: C**
Ammonia is extremely poisonous to cells and must be heavily diluted with water to prevent tissue damage during its removal.
2. **Correct Answer: B**
Terrestrial animals convert ammonia into urea because it can be safely stored in the body for short periods and requires much less water for excretion.
3. **Correct Answer: C**
While vertebrates have complex kidneys, most invertebrates utilize simpler tubular systems like protonephridia, metanephridia, or Malpighian tubules.
4. **Correct Answer: B**
Plants lack specialized excretory organs and instead store metabolic wastes in cellular vacuoles or in leaves that eventually fall off.
5. **Correct Answer: C**
This medical procedure, also known as dialysis, is used to artificially filter the blood when the natural kidneys are no longer functional.
6. **Correct Answer: D**
This unique tissue lines the ureters and bladder, allowing these organs to stretch and expand as they fill with urine.

7. **Correct Answer: A**
   Insects, such as cockroaches and grasshoppers, use Malpighian tubules to remove nitrogenous wastes from their body fluid (hemolymph).
8. **Correct Answer: B**
   Sweat glands primarily function to regulate body temperature through evaporation while simultaneously eliminating small amounts of salts and urea.
9. **Correct Answer: B**
   The lungs serve as vital excretory organs for the removal of gaseous CO_2 produced as a byproduct of cellular respiration.
10. **Correct Answer: C**
    The main purpose of an artificial kidney is to filter out harmful nitrogenous substances like urea and uric acid from the patient's bloodstream.

ONE OR MORE THAN ONE CORRECT OPTION TYPE

11. **Correct Answer: A, B, C**
    These three processes (filtration, reabsorption, and secretion) work sequentially in the nephron to filter blood and form final urine.
12. **Correct Answer: A, B, C**
    The lungs (CO_2), liver (bile pigments/urea), and skin (salts) are accessory organs that assist the kidneys in maintaining a clean internal environment.
13. **Correct Answer: A, B, C**
    Micturition (urination) occurs when stretch receptors trigger a reflex that contracts the detrusor muscle via parasympathetic nervous signals.
14. **Correct Answer: B, C, D**
    Plants do not have specialized organs; they release CO_2 at night, store wastes in vacuoles/leaves, and use transpiration to expel excess water.
15. **Correct Answer: A, B, C**
    ADH, Aldosterone, and Angiotensin II are the primary hormones that manage water and salt balance to regulate blood pressure and fluid levels.

ASSERTION AND REASON TYPE

16. **Correct Answer: B**
    While both statements correctly describe the anatomical placement of nephron parts, the location of the loop of Henle does not explain why the other components are situated in the cortex.
17. **Correct Answer: B**
    Both definitions are scientifically accurate, but the process of defecation (eliminating undigested food) is a distinct biological process that does not explain what metabolic excretion is.
18. **Correct Answer: D**
    Glomerular filtration is a passive process that does not require cellular energy (ATP); it is driven entirely by the hydrostatic pressure of the blood in the capillaries.
19. **Correct Answer: A**
    The collecting duct is responsible for final water recovery because Vasopressin (ADH) creates "water channels" (aquaporins) that allow water to move back into the body.

MATRIX MATCHING TYPE

20. **Correct Answer: A**
    a → S, b → Q, c → R, d → P
21. **Correct Answer: A**
    a → R, b → Q, c → P, d → S

COMPREHENSION TYPE

Passage1.

22. **Correct Answer: C**
    Ammonia is extremely toxic and highly soluble, requiring approximately 300–500 ml of water per gram of nitrogen excreted to prevent tissue poisoning.
23. **Correct Answer: B**
    While the kidney *removes* urea from the blood, it is actually the liver that produces it by converting toxic ammonia via the Ornithine (Urea) cycle.
24. **Correct Answer: C**
    Uric acid is the least toxic and least soluble waste; animals like birds and desert reptiles excrete it as a paste with almost no water loss.

Passage2.

25. Correct Answer: C

Urine formation begins with the passive movement of water and solutes from the blood into the Bowman's capsule due to high hydrostatic pressure.

26. Correct Answer: C

Tubular secretion removes ions like H^+ and K^+ from the blood into the tubule, which is critical for maintaining the body's pH and ionic stability.

27. Correct Answer: C

The Proximal Convoluted Tubule (PCT) is lined with brush border epithelium that reabsorbs nearly 70-80% of electrolytes, water, and all essential nutrients like glucose.

Previous Year Questions

1. Correct Answer: C

The breakdown of hemoglobin in the liver produces a pigment called urochrome (or urobilin), which gives urine its characteristic straw-yellow color.

2. Correct Answer: C

Since uric acid is nearly insoluble in water, it is excreted as a semisolid paste or pellet to maximize water conservation in birds and reptiles.

3. Correct Answer: D

In the standard diagram of a Malpighian corpuscle, the Afferent arteriole (wider) enters, the Efferent (narrower) leaves, and the tubule shown leading away from the capsule area is the Distal Convoluted Tubule (DCT) or PCT depending on the specific loop depicted.

4. Correct Answer: C

These groups are uricotelic, meaning they convert nitrogenous waste into uric acid to survive in environments where water is scarce or to stay light for flight.

5. Correct Answer: C

The hilum is the notch or "gateway" on the inner concave surface of the kidney where the renal artery, renal vein, nerves, and ureter pass through.

6. Correct Answer: D

The hilum is the notch or "gateway" on the inner concave surface of the kidney where the renal artery, renal vein, nerves, and ureter pass through.

7. Correct Answer: D

The hilum is the notch or "gateway" on the inner concave surface of the kidney where the renal artery, renal vein, nerves, and ureter pass through.

8. Correct Answer: C

The hilum is the notch or "gateway" on the inner concave surface of the kidney where the renal artery, renal vein, nerves, and ureter pass through.

9. Correct Answer: C

Because urea is synthesized in the liver through the Ornithine cycle, the hepatic vein (which carries blood away from the liver) contains the highest concentration of urea before it is diluted.

10. Correct Answer: C

Tubular secretion of H^+ ions into the renal tubules lowers the pH of the filtrate, typically making human urine slightly acidic (average pH 6.0).

5. Control and coordination

Solutions

Worksheet – 1

1. **Correct Answer: C Neuron**

   Neurons are the specialized cells that transmit electrical impulses. While "nerves" are bundles of these neurons, the individual neuron is the fundamental unit.

2. **Correct Answer: C Axon**

   The **axon** is the long projection that carries impulses away from the cell body. It ends in "axon terminals," which are the branched distal ends.

3. **Correct Answer: B Brain and spinal cord**

   The CNS acts as the command center, consisting of the brain and the spinal cord. Everything else (nerves branching out) belongs to the Peripheral Nervous System (PNS).

4. **Correct Answer: B Cerebral cortex**
   The cerebral cortex is the outer layer of the cerebrum, composed primarily of neuron cell bodies, which gives it that distinctive grey color.
5. **Correct Answer: D Subarachnoid space**
   The meninges consist of the dura mater (outer), arachnoid (middle), and pia mater (inner). The space below the arachnoid is the subarachnoid space, which contains cerebrospinal fluid (CSF).
6. **Correct Answer: B Thalamus/hypothalamus and pons**
   The midbrain serves as a bridge between the forebrain (specifically the diencephalon, which includes the thalamus/hypothalamus) and the hindbrain (starting with the pons).
7. **Correct Answer: C Little brain**

   In Latin, "cerebellum" literally translates to "little brain." It looks like a miniature version of the cerebrum tucked under the back of the head.
8. **Correct Answer: C Spinal cord**
   The **medulla oblongata** is the lowest part of the brainstem and passes through the foramen magnum of the skull to become the spinal cord.
9. **Correct Answer: B Midbrain, pons, medulla oblongata**
   These three structures together form the brainstem, which controls many of the body's involuntary life-support functions**.**
10. **Correct Answer: C Corpora quadrigemina**
    The dorsal (back) part of the midbrain consists of four rounded lobes called the corpora quadrigemina, which are involved in visual and auditory reflexes**.**

Worksheet-2

1. **Correct Answer: C Brain and spinal cord**
   The PNS includes all nerves (cranial and spinal) outside the brain and spinal cord.
2. **Correct Answer: C 12 pairs**
   Humans have 12 pairs of cranial nerves and 31 pairs of spinal nerves.
3. **Correct Answer: C Dorsal and ventral roots**
   Spinal nerves are "mixed" nerves formed by the union of the dorsal (sensory) root and ventral (motor) root.
4. **Correct Answer: C Both sensory and motor**
   The Somatic system relays impulses from CNS to skeletal muscles (motor) and carries sensory info to the CNS.
5. **Correct Answer: C Craniosacral division**
   It is called the Craniosacral division because its fibers emerge from the brainstem and the sacral region of the spinal cord.
6. **Correct Answer: B Automatic and involuntary)**
   Reflexes are designed for speed and survival, meaning they are automatic, involuntary, and instantaneous.

7. **Correct Answer: B Bloodstream or lymph**
Endocrine glands lack ducts and must secrete hormones directly into the bloodstream or lymph**.**
8. **Correct Answer: C Single amino acid**
Amino hormones (like adrenaline or thyroxine) are typically derivatives of a single amino acid (usually tyrosine or tryptophan).
9. **Correct Answer: B Endocrine glands**
Endocrine glands are ductless; exocrine glands (like sweat glands) use ducts.
10. **Correct Answer: B Peptide hormone**
Vasopressin (ADH) and Oxytocin are both classified as peptide hormones (specifically nonapeptides).

Worksheet-3

1. **Correct Answer: C) Diencephalon**
This part of the forebrain consists of the thalamus and the hypothalamus**.**
2. **Correct Answer: C) Seminiferous tubules**
These are the specific locations within the testes where sperm production (spermatogenesis) occurs**.**
3. **Correct Answer: C) Neurohypophysis**
The pituitary gland is divided into the adenohypophysis (anterior) and the neurohypophysis (posterior).
4. **Correct Answer: B) Oestrogen and progesterone**
These are the primary steroid hormones produced by the ovaries to regulate the female reproductive cycle.
5. **Correct Answer: C) Mental retardation and stunted growth**
Cretinism is caused by severe hypothyroidism (thyroid deficiency) during infancy or pregnancy**.**
6. **Correct Answer: B) Loss of glucose in urin**
This condition is known as glycosuria, a hallmark sign of Diabetes mellitus due to high blood sugar levels.
7. **Correct Answer: C) Sleep–wake cycle**
Melatonin is secreted by the pineal gland and helps regulate our circadian rhythm.
8. **Correct Answer: C Weak immune response**
The thymus gland produces thymosins for T-cell maturation; as it shrinks with age, immunity generally weakens.
9. **Correct Answer: C Suprarenal glands**
This name literally means "above the kidney," which is exactly where the adrenal glands are located.
10. **Correct Answer: C Carbohydrates**
While they do affect fats and proteins, glucocorticoids (like cortisol) are primarily known for stimulating gluconeogenesis**.**

Competitive Corner

SINGLE CORRECT OPTION TYPE

1. **Correct Answer: C Neural and endocrine systems together**
The National Institutes of Health (NIH) describes how the nervous system provides rapid, short-lived coordination, while the endocrine system provides slower, long-lasting chemical integration**.**
2. **Correct Answer: C Mixed neurons**
Mixed neurons (or mixed nerves) contain both sensory fibers (carrying impulses to the CNS) and motor fibers (carrying impulses away from the CNS).
3. **Correct Answer: B Infundibulum**
According to Britannica, the infundibulum is the stalk-like structure that attaches the pituitary gland to the hypothalamus.
4. **Correct Answer: B Shock absorber**
CSF cushions the brain and spinal cord within the bony skull and vertebral column to prevent mechanical injury.
5. **Correct Answer: C Dorsal and ventral roots**
Each spinal nerve is formed by the junction of a dorsal (sensory) root and a ventral (motor) root.
6. **Correct Answer: C Voluntary movements**
The somatic nervous system (SNS) carries signals to skeletal muscles for conscious control.
7. **Correct Answer: C Craniosacral division**
    - Parasympathetic fibers emerge from the cranial nerves and the sacral segments of the spinal **cord.**

8. **Correct Answer: C Growth hormone**
   - GH is synthesized and released by the anterior pituitary (adenohypophysis**).**

9. **Correct Answer: B β-cells**

   The American Diabetes Association notes that beta cells in the Islets of Langerhans produce insulin.

10. **Correct Answer: C Catecholamines**

    These "fight or flight" hormones are structurally classified as catecholamines

11. **Correct Answer: C Gigantism**

    Excess GH before the epiphyseal plates (growth plates) close results in abnormal height.

12. **Correct Answer: C Libido**

    In biological and medical terms, libido refers to the overall sexual drive or behavior influenced by hormones like testosterone.

ONE OR MORE THAN ONE CORRECT OPTION TYPE

13. **Correct Answer: A, B, and D**
    A. Thermoregulation: It acts as the body's thermostat.
    B. Thirst and hunger control: It contains the satiety and thirst centers.
    C. Hormone secretion: It produces releasing hormones and neurohormones (like ADH and Oxytocin) via neurosecretory cells.

14. **Correct Answer: A, B, and C**

    A) Glucocorticoids: e.g., Cortisol (involved in glucose metabolism).B) Mineralocorticoids: e.g., Aldosterone (involved in salt/water balance).C) Androgens: Small amounts of sex hormones are produced here.

15. **Correct Answer: A, B, and D**

    A) Pupils: Sympathetic (fight or flight) dilates; Parasympathetic (rest and digest) constricts.

    B) Heartbeat: Sympathetic increases rate; Parasympathetic slows it down.

    D) Bronchi: Sympathetic relaxes them (to breathe more during stress); Parasympathetic constricts them.

16. **Correct Answer: A, C, and D**

    A) Automatic and involuntary

    C) Instantaneous and unconscious**:** It happens before the brain fully processes the sensation.

    D) **Involves reflex arc** is the structural requirement for the action to occur.

17. **Correct Answer: A, B, and D**

    A) Prolonged hyperglycemia: High blood sugar levels are the primary symptom.

    B) Loss of glucose through urine: Known as glycosuria.

    D) Formation of ketone bodies: In severe cases (Type 1), the body breaks down fats for fuel, creating ketones.

ASSERTION AND REASON TYPE

18. **Correct Answer: A**

    Both A and R are true, and R is the correct explanation of A

19. **Correct Answer: C**

    A is true, but R is false

20. **Correct Answer: B**

    Both A and R are true, but R is not the correct explanation of A

21. **Correct Answer: C**

    A is true, but R is false

MATRIX MATCHING TYPE

22. **Correct Answer: A**

    a → Q, b → P, c → R, d → S

23. **Correct Answer: A**

    a → P, b → Q, c → R, d → S

COMPREHENSION TYPE

PASSAGE-01

24. Correct Answer: C

Neuron

25. Correct Answer: C

Spinal cord

26. Correct Answer: D

Peripheral nervous system

PASSAGE-2

27. Correct Answer: C

chemical messengers

28. Correct Answer: C

Cretinism

29. Correct Answer: C

Hypothalamus

PREVIOUS YEAR QUESTIONS

1. Correct Answer: C

Reason:

- Statement (i) correctly defines a reflex action. Statement (ii) is incorrect because motor neurons (not sensory) carry signals from the spinal cord to muscles. Statement (iii) is incorrect because sensory neurons (not motor) carry signals from receptors to the spinal cord.

2. Correct Answer: C Pancreas — Differentiation of T-Lymphocytes of immune system

Reason:

This is an incorrect match. The Thymus gland is responsible for the differentiation of T-Lymphocytes, not the pancreas. The pancreas primarily regulates blood sugar.

3. Correct Answer: B (i) and (iii) only

(i) Correctly describes the path from dendrite to axon end. (iii) Correctly states that impulses are **unidirectional** due to the location of neurotransmitters.

4. Correct Answer: B Acetylcholine

Acetylcholine is a primary neurotransmitter used to conduct signals across the synaptic cleft between neurons.

5. Correct Answer: C One afferent and one efferent neuron

A basic reflex arc requires at least one afferent (sensory) neuron to bring the signal in and one efferent (motor) neuron to send the response out.

6. Correct Answer: D I, II and III

All statements are correct.
The midbrain is located between the forebrain and hindbrain, contains the **cerebral aqueduct**, and its dorsal part consists of the **corpora quadrigemina** (four lobes).

7. Correct Answer: C Neural and endocrine system

These two systems work together (neuro-endocrine coordination) to maintain homeostasis and integrate all body activities.

8. Correct Answer: C Reflex action

A reflex action is defined as an involuntary, rapid response to a peripheral stimulus.

9. Correct Answer: B Autonomic neural system

The Autonomic Nervous System (ANS) is divided into the sympathetic and parasympathetic divisions.

10. Correct Answer: D Excitability and conductivity

Neurons are characterized by their ability to be stimulated (excitability) and their ability to transmit that stimulus along their length (conductivity).

6. REPRODUCTION

Solutions

Worksheet – 1

1. **Correct Answer: B) Errors during DNA copying**
   Explanation: Variations arise mainly due to errors during DNA replication, which can lead to mutations and genetic diversity.
2. **Correct Answer: B) Variations**
   Explanation: Variations provide the raw material for evolution; they allow natural selection to act and shape populations over time.
3. **Correct Answer: B) Asexual**
   Explanation: Fission is a form of asexual reproduction where a single organism divides to form new individuals.
4. **Correct Answer: C) Amoeba**
   Explanation: Simple binary fission occurs in Amoeba, where the nucleus divides followed by the cytoplasm.
5. **Correct Answer: D) Plasmodium**
   Explanation: Plasmodium exhibits multiple fission (schizogony) in its life cycle, producing many daughter cells at once.
6. **Correct Answer: C) Paramecium**
   Explanation: Transverse binary fission is typical of Paramecium, where the cell divides across its short axis.
7. **Correct Answer: D) Leishmania**
   Explanation: Longitudinal binary fission occurs in Leishmania, where the cell divides along its long axis.
8. **Correct Answer: C) Entire body of the organism**
   Explanation: In fission, the whole organism acts as the reproductive unit and divides to form offspring.
9. **Correct Answer: D) Transverse fission**
   Explanation: Leishmania, the causative agent of kala-azar, reproduces by transverse binary fission.
10. **Correct Answer: B) Two**
    Explanation: Binary fission produces two genetically identical offspring from one parent organism.

Worksheet-2

1. **Correct Answer: B) Multicellular organisms**
   Explanation: Fragmentation occurs in multicellular organisms where the body breaks into fragments, each capable of forming a new organism.
2. **Correct Answer: C) Pseudo mycelium**
   Explanation: In yeast, buds that remain attached form a chain-like structure called pseudo mycelium.
3. **Correct Answer: C) Buds**
   Explanation: Budding is an asexual process where a new individual develops from a small outgrowth or bud on the parent organism.
4. **Correct Answer: C) Regrow lost body parts**
   Explanation: Regeneration is the ability of an organism to regrow lost or damaged body parts.
5. **Correct Answer: C) Self-amputation**
   Explanation: Autotomy is the deliberate shedding of a body part, usually as a defensive mechanism (e.g., lizard tail).
6. **Correct Answer: B) Asexual reproduction**
   Explanation: Vegetative propagation is an asexual method where new plants develop from vegetative parts like stems, roots, or leaves.
7. **Correct Answer: B) Bryophyllum**
   Explanation: Bryophyllum produces adventitious buds along its leaf margins, which can grow into new plants.
8. **Correct Answer: B) Sporangia**
   Explanation: In Rhizopus, spores are produced inside sporangia, which are specialized structures at the tips of hyphae.

9. **Correct Answer: D) Grafting**
Explanation: Grafting is an artificial method of vegetative propagation, unlike runners, rhizomes, and tubers, which are natural methods.
10. **Correct Answer: B) Simple multicellular organisms**
Explanation: Spore formation is common in simple multicellular organisms like fungi, algae, and bryophytes, not in higher plants or animals.

Worksheet-3

1. **Correct Answer: B) Two parents**
Explanation: Sexual reproduction requires two parents to contribute gametes (male and female) for the formation of offspring.
2. **Correct Answer: C) Meiosis**
Explanation: Gametes are produced by meiosis, which reduces the chromosome number by half to maintain the species' chromosome count after fertilization.
3. **Correct Answer: C) Isogamy**
Explanation: Isogamy is the fusion of gametes that are morphologically and physiologically identical.
4. **Correct Answer: B) Bisexual or hermaphrodites**
Explanation: Organisms with both male and female reproductive organs are called hermaphrodites or bisexual.
5. **Correct Answer: B) Calyx and corolla**
Explanation: Calyx (sepals) and corolla (petals) are accessory whorls that do not directly participate in reproduction.
6. **Correct Answer: D) Flower**
Explanation: In sexually reproducing plants, the flower is the reproductive unit as it contains the male and female reproductive organs.
7. **Correct Answer: C) Fertilization**
Explanation: Fertilization restores the diploid chromosome number by fusing haploid male and female gametes.
8. **Correct Answer: B) Fertilization**
Explanation: Fertilization is the union of male and female gametes to form a zygote.
9. **Correct Answer: C) Participate directly in reproduction**
Explanation: Essential whorls (androecium and gynoecium) are directly involved in reproduction.
10. **Correct Answer: C) Female reproductive organs**
Explanation: The gynoecium collectively represents the female reproductive organs of a flower.

Worksheet-4

1. **Correct Answer: C) Seminiferous tubules**
Explanation: Spermatogenesis, the production of sperm, occurs in the seminiferous tubules of the testes.
2. **Correct Answer: C) Storage and maturation of sperms**
Explanation: The epididymis stores sperm and allows them to mature before ejaculation.
3. **Correct Answer: C) Bulbourethral glands**
Explanation: Cowper's glands are also called bulbourethral glands; they secrete a lubricating fluid.
4. **Correct Answer: C) Bulbourethral gland**
Explanation: The bulbourethral gland secretes a mucus-like fluid that lubricates the penis during sexual activity.
5. **Correct Answer: B) Ovaries, oviducts, uterus, vagina, and external genitalia**
Explanation: The female reproductive system includes internal organs (ovaries, oviducts, uterus, vagina) and external genitalia.
6. **Correct Answer: C) Womb**
Explanation: The uterus is commonly called the womb, where the fertilized egg develops into a fetus.
7. **Correct Answer: B) Cervix to vestibule**
Explanation: The vagina extends from the cervix of the uterus to the vestibule of the external genitalia.
8. **Correct Answer: B) External genitalia of the female**
Explanation: The vulva or pudendum refers to the external genital organs of the female.

9. **Correct Answer: C) Primary sex organs**
Explanation: Gametogenesis occurs in the primary sex organs (testes in males, ovaries in females).
10. **Correct Answer: C) Gametes**
Explanation: Gametogenesis is the process of formation of gametes (sperm in males, eggs in females).

Competitive Corner

SINGLE CORRECT OPTION TYPE

1. **Correct Answer: B) To help offspring survive in a specific ecological niche**
Explanation: DNA replication is stable to ensure genetic information is accurately passed on, helping offspring survive in their environment.
2. **Correct Answer: C) DNA copying**
Explanation: Variations arise mainly due to occasional errors during DNA replication (copying), leading to mutations.
3. **Correct Answer: C) Number of nuclear divisions**
Explanation: Fission is classified into simple, transverse, longitudinal, or multiple fission based on how many times the nucleus divides.
4. **Correct Answer: C) Many daughter cells simultaneously**
Explanation: Multiple fission produces many daughter cells at once, as seen in Plasmodium.
5. **Correct Answer: B) Follicle-Stimulating Hormone (FSH)**
Explanation: Ovaries produce ovum, estrogen, and progesterone; FSH is secreted by the pituitary gland.
6. **Correct Answer: C) Ampulla**
Explanation: The ampulla is the widest part of the oviduct, where fertilization usually occurs.
7. **Correct Answer: C) Erectile tissue**
Explanation: The penis contains erectile tissue that fills with blood to enable erection.
8. **Correct Answer: C) Highly specialized tissues and organs**
Explanation: Fragmentation requires simple body organization; highly specialized organisms cannot regenerate from fragments easily.
9. **Correct Answer: C) Autotomy**
Explanation: Autotomy is the self-amputation of a body part (e.g., lizard tail) as a defense mechanism.
10. **Correct Answer: C) Sporangia**
Explanation: In Rhizopus, the sporangia are the reproductive structures that produce spores.

ONE OR MORE THAN ONE CORRECT OPTION TYPE

11. **Correct Answer: A) Single parent involved, B) Rapid multiplication, C) No fusion of gametes**
Explanation: Asexual reproduction involves a single parent, allows rapid multiplication, and does not involve gamete fusion. It produces low genetic variation, not high.
12. **Correct Answer: A, B, C**
Budding is observed in
    - **Yeast: TRUE.** (Unicellular budding).
    - **Hydra: TRUE.** (Multicellular budding).
    - **Sponges: TRUE.** (Internal and external budding).
13. **Correct Answer: A) Sepals, B) Petals**
Explanation: Sepals and petals are non-reproductive (accessory) parts of a flower; stamens and carpels are reproductive.
14. **Correct Answer: A) Ovaries, B) Oviducts, C) Uterus**
Explanation: The female reproductive system includes ovaries, oviducts (fallopian tubes), and uterus. Vas deferens is part of the male reproductive system.

15. **Correct Answer: A) Is an androgen, B) Is secreted by Leydig cells, C) Regulates male reproductive functions**

Explanation: Testosterone is an androgen hormone secreted by Leydig cells in the testes and regulates male reproductive functions. It is not produced in seminiferous tubules (which produce sperm).

ASSERTION AND REASON TYPE

16. **Correct Answer: A) Both Assertion and Reason are true, and Reason correctly explains Assertion**

Explanation: Variations allow some individuals to survive sudden environmental changes, which is essential for the long-term survival of a species.

17. **Correct Answer: B) Both Assertion and Reason are true, but Reason does not explain Assertion**

Explanation: Amoeba lacks a fixed plane of division, and it undergoes simple binary fission, but the type of fission does not explain the lack of a fixed plane.

18. **Correct Answer: A) Both Assertion and Reason are true, and Reason correctly explains Assertion**

Explanation: In Planaria, regeneration allows each body fragment to develop into a complete organism, serving as a method of asexual reproduction.

19. **Correct Answer: A) Both Assertion and Reason are true, and Reason correctly explains Assertion**

Explanation: Epididymis is important for male fertility because it stores and allows sperm to mature temporarily.

MATRIX MATCHING TYPE

20. **Correct Answer: A)**

a–>Q, b–>P, c–>R, d–>S

21. **Correct Answer: A)**

a–>P, b–>Q, c–>R, d–>S

COMPREHENSION TYPE

(PASSAGE-1)

22. **Correct Answer: B) producing new individuals**

Explanation: Reproduction is the process of producing new individuals.

23. **Correct Answer: C) variations**

Explanation: Small differences during DNA copying are called variations.

24. **Correct Answer: B) helps organisms survive changes**

Explanation: Variation is useful because it provides a survival advantage in changing environments.

PASSAGE-2

25. **Correct Answer: B) Calyx and corolla**

Explanation: Calyx (sepals) and corolla (petals) are non-reproductive whorls.

26. **Correct Answer: B) ovule**

Explanation: The ovule develops into a seed after fertilization.

27. **Correct Answer: C) androecium**

Explanation: The androecium produces male gametes (pollen) in flowering plants.

PREVIOUS YEAR QUESTIONS

1. **Correct Answer: A) T F T F**

Explanation:

i. True – Umbilical cord links embryo to placenta.
ii. False – Gastrulation is the formation of three germ layers from the blastula.
iii. True – Scrotum regulates temperature for sperms.
iv. False – Progesterone is secreted by the ovary, not anterior pituitary.

2. **Correct Answer: A) Colocasia of group I must be interchanged with sweet potato of Group III.**

Explanation: Colocasia is propagated by tubers, not leaves; sweet potato belongs to tuberous group.

3. **Correct Answer: C) (ii) and (iii) only**

Explanation: Seminal vesicle provides energy to sperm and helps transport; it does not secrete FSH.

4. **Correct Answer: A) Both statements 1 and 2 are true and statement 2 is the correct explanation of statement 1**
   Explanation: Oral contraceptives are hormone-containing tablets used to prevent pregnancy.
5. **Correct Answer: D) In a normal human female, implantation of the blastocyst to the uterine wall occurs after 17 days of fertilisation.**
   Explanation: Normal implantation occurs about 6–7 days after fertilization, not 17.
6. **Correct Answer: C) Reversible – Tubal ligation reversibility is very poor**
   Explanation: Contraceptive pills are reversible; tubal ligation is mostly irreversible.
7. **Correct Answer: B) 1 and 3**
   Explanation: Plant Y has larger, heavier pollen → more pollen must be produced for effective pollination. Its spiky, sticky pollen suggests insect pollination.
8. **Correct Answer: A) Salpa → Plasmodium → Hydra**
   Explanation: Salpa (simple fission), Plasmodium (multiple fission), Hydra (budding).
9. **Correct Answer: C) Vegetative propagation by stem – garlic**
   Explanation: Garlic is propagated by modified underground stem (cloves), not just stem in general.
10. **Correct Answer: C) (i), (ii) and (iv) only**
    Explanation: Vegetative propagation allows seedless plant propagation, produces genetically identical offspring, and tissue culture ensures disease-free plants. It does not improve adaptability or dispersal.
    I can also make a complete solved worksheet PDF with all answers and explanations if you want—it will be neatly formatted for quick revision.
    Do you want me to do that?

7. Heredity and Evolution

Solutions

Worksheet – 1

1. **Correct Answer: B**
   Mendel is called the Father of Genetics because he discovered the fundamental laws of inheritance through his work with pea plants.
2. **Correct Answer: B**
   He chose the garden pea because it was easy to grow, had a short life cycle, and possessed clear, contrasting traits.
3. **Correct Answer: C**
   Mendel specifically tracked seven pairs of contrasting characters, such as tall vs. dwarf stems and round vs. wrinkled seeds.
4. **Correct Answer: D**
   In a monohybrid cross, the F_2 generation typically shows three individuals with the dominant trait for every one with the recessive trait.
5. **Correct Answer: B**
   The prefix "mono-" means one; this cross focuses specifically on the inheritance of a single characteristic.
6. **Correct Answer: A**
   The 9:3:3:1 ratio represents the four possible phenotype combinations when two different traits are tracked simultaneously.
7. **Correct Answer: B**
   Mendel was an Augustinian friar (monk) who lived and conducted his famous experiments in the monastery garden in Brno.
8. **Correct Answer: C**
   Since plant height is a single trait (tall vs. dwarf), studying it alone constitutes a monohybrid cross.
9. **Correct Answer: B**
   A dihybrid cross involves two traits (e.g., seed color and seed shape) to see if they are inherited independently of each other.
10. **Correct Answer: D**
    In his famous dihybrid experiment, Mendel specifically used Seed Shape (round/wrinkled) and Seed Colour

Worksheet – 2

1. **Correct Answer: C**
   Human somatic (body) cells contain 23 pairs of chromosomes, totaling 46.
2. **Correct Answer: B**
   Females are XX, so every egg they produce receives one X chromosome, making them homogametic.
3. **Correct Answer: C**
   Beyond deciding if an individual is male or female, these chromosomes carry genes for traits like color vision and blood clotting.
4. **Correct Answer: B**
   Since mothers always give an X, the child's sex depends on whether the father's sperm carries an X (for a girl) or a Y (for a boy).
5. **Correct Answer: B**
   The biological sex is permanently established at the moment the sperm's sex chromosome (X or Y) combines with the egg's X chromosome.
6. **Correct Answer: C**
   In humans, the XX combination results in a female, while XY results in a male.
7. **Correct Answer: C**

   It is often called this because it applies to all sexually reproducing organisms without exception.
8. **Correct Answer: B**

   This term specifically refers to the genetic mechanism that establishes the biological sex of an organism.
9. **Correct Answer: D**

   While Mendel founded genetics, the Law of Evolution (Natural Selection) was proposed by Charles Darwin.

10. **Correct Answer: B**
Because sperm can be either X or Y, it is the male gamete that ultimately determines the sex of the offspring.

Worksheet-3

1. **Correct Answer: C**
He is considered the Father of Modern Embryology for discovering the mammalian ovum and establishing the "von Baer's laws" of development.
2. **Correct Answer: B**
These organs share a common evolutionary origin and skeletal structure, even if they perform different tasks (like a human arm and a bird's wing).
3. **Correct Answer: C**
These organs serve the same purpose but evolved independently from different structures, such as the wings of a butterfly versus those of a bat.
4. **Correct Answer: A**
Peripatus (the velvet worm) is a "living fossil" that displays segmented features of annelids and the jointed legs and tracheal system of arthropods.
5. **Correct Answer: D**
These are evolutionary "leftovers," like the human appendix or wisdom teeth, that have lost their original utility over time.
6. **Correct Answer: B**
Paleontology is the scientific study of life in past geologic periods through the analysis of plant and animal fossils.
7. **Correct Answer: A**
Both share the same pentadactyl (five-fingered) bone arrangement, proving they descended from a common mammalian ancestor.
8. **Correct Answer: C**
Atavism refers to the occasional reappearance of an ancestral trait, such as a human baby born with a small vestigial tail.
9. **Correct Answer: B**
Archaeopteryx is the famous link between reptiles and birds, while Cynognathus shows features transitioning from reptiles to mammals.
10. **Correct Answer: D**
Darwin proposed that individuals with traits better suited to their environment are more likely to survive and reproduce, a process he called natural selection.

Worksheet – 4

1. **Correct Answer: C**
He proposed the first coherent theory of evolution, focusing on the inheritance of acquired characteristics and the law of use and disuse.
2. **Correct Answer: B**
Darwinism is based on his observations and the theory of evolution by natural selection as detailed in his 1859 book.
3. **Correct Answer: D**
These are traits gained through environmental influence or use/disuse during an organism's life, rather than being coded in the DNA from birth.
4. **Correct Answer: B**
Darwin's observations of diverse flora and fauna during this voyage, particularly in the Galápagos Islands, formed the basis of his evolutionary ideas.
5. **Correct Answer: C**
This concept belongs to Lamarckism; Darwinism specifically rejected the idea that physical changes gained during life could be passed to offspring.
6. **Correct Answer: B**
Hugo de Vries proposed that evolution occurs through sudden, large changes (mutations) rather than the slow, gradual process Darwin envisioned.
7. **Correct Answer: D**
This is a "trick" choice because Mendel did not write *Origin of Species* (Darwin did) and Mendel's work was largely unknown to Darwin during his lifetime.
8. **Correct Answer: B**
Malthus's essay on population growth and competition for resources helped Darwin formulate his idea of the "struggle for existence."

9. **Correct Answer: B**
The core of Darwin's theory is that nature "selects" the most adaptable individuals to survive and pass on their traits.
10. **Correct Answer: C**
Lyell's work on uniformitarianism taught Darwin that the Earth was much older than previously thought, providing the time needed for slow evolution.

Competitive Corner

SINGLE CORRECT OPTION TYPE

1. **Correct Answer: B**
Mendel is the scientist who discovered the fundamental principles of heredity through his systematic breeding of pea plants.
2. **Correct Answer: C**
Mendel was born in July 1822 and passed away in January 1884, with his most famous work being published in 1866.
3. **Correct Answer: B**
Humans have 22 pairs of non-sex chromosomes (autosomes) and one pair of sex chromosomes (XX or XY) for a total of 46.
4. **Correct Answer: D**
This law states that allele pairs separate during gamete formation, so each gamete carries only one allele for each gene.
5. **Correct Answer: B**
Darwin's theory of natural selection focuses on variation, the struggle for existence, and survival of the fittest, rather than migration patterns.
6. **Correct Answer: A**
Homology refers to structures in different species that are similar because of common ancestry, despite having different functions.
7. **Correct Answer: A**
Both structures are modified axillary buds, meaning they share the same anatomical origin even though one is for protection and the other for climbing.
8. **Correct Answer: B**
While Lamarck famously used the giraffe as an example of acquired traits, modern science views the long neck as a result of natural selection favoring those who could reach higher food.
9. **Correct Answer: C**
These are remnants of limbs from evolutionary ancestors that are no longer functional in modern pythons or whales.
10. **Correct Answer: A**
In the human XY sex-determination system, the presence of the Y chromosome triggers the development of male biological characteristics.

ONE OR MORE THAN ONE CORRECT OPTION TYPE

11. **Correct Answer: A, B, C**

Mendel chose pea plants because they grow quickly, have easily observable opposite traits, and allow for controlled pollination.

12. **Correct Answer: A, B, D**

Wallace was a naturalist in the Malay Archipelago who independently arrived at the same conclusions about natural selection as Darwin.

13. **Correct Answer: A, B, C, D**

Darwin's theory relies on the observation that species overproduce, populations remain stable, and variations lead to a struggle for survival.

14. **Correct Answer: A, B, C**

The vermiform appendix, ear pinna muscles, and body hair are all structures that have lost their original function in humans.

15. **Correct Answer: A, B, C**

In these crosses, the F_1 shows only the dominant trait, while the F_2 splits into a 3:1 ratio because alleles separate during gamete formation.

ASSERTION AND REASON TYPE

16. **Correct Answer: A**

Reproductive isolation is essential for speciation because it prevents different groups from interbreeding and merging back into a single population.

17. **Correct Answer: A**
While life currently comes from life (Biogenesis), the scientific study of the "origin of life" seeks to explain the unique event where the first living structures initially formed.
18. **Correct Answer: A**
Mendel's groundbreaking work was published in an obscure journal and used complex mathematics that the biologists of his time were not yet ready to understand or accept.
19. **Correct Answer: A**
Pasteur's famous swan-neck flask experiment proved that microorganisms only appear in sterile broth when exposed to existing cells, effectively killing the idea of spontaneous generation.

MATRIX MATCHING TYPE

20. **Correct Answer: A**
a → P, b → Q, c → R, d → S
21. **Correct Answer: A**
a → P, b → Q, c → R, d → S

COMPREHENSION TYPE

Passage1

22. **Correct Answer: B**
In a typical dihybrid cross, Mendel crossed pure-breeding plants with two pairs of contrasting traits to observe their inheritance.
23. **Correct Answer: C**
Due to the Law of Dominance, the F_1 generation only displays the dominant traits, which in this case are round shape and yellow color.
24. **Correct answer –D**

Self-pollination of the F_1 plants leads to four different seed combinations in the F_2 generation: Round-Yellow, Round-Green, Wrinkled-Yellow, and Wrinkled-Green.

Passage2

25. **Correct Answer: B**

Palaeontology is the scientific field dedicated to studying prehistoric life forms by examining fossils preserved in the Earth's crust.

26. **Correct answer –C**

This branch of biology focuses on the formation and development of embryos, revealing evolutionary links through similar growth stages in different species.

27. **Correct Answer: C**

By comparing the physical structures of different species, scientists can determine how closely they are related and identify common ancestors.

Previous Year QUESTIONS

1. **Correct Answer: A**

Chrysemys picta (the painted turtle) exhibits Temperature-dependent Sex Determination (TSD), where the incubation temperature of the eggs determines the sex of the offspring, unlike humans who use chromosomes.

2. **Correct Answer: C**

In a dihybrid cross involving two heterozygous parents (RrYy \times RrYy), the independent assortment of alleles results in four phenotypes in a predictable 9 (Dominant/Dominant): 3 (Dominant/Recessive): 3 (Recessive/Dominant): 1 (Recessive/Recessive) ratio.

3. **Correct Answer: D**

In Mendel's pea plant experiments, Yellow seed color is dominant over Green, meaning the green trait only appears if the plant is homozygous recessive (yy).

4. **Correct Answer: A**

The F_1 generation will have the genotype TtYy. Because "T" (Tall) is dominant over "t" (dwarf) and "Y" (Yellow) is dominant over "y" (green), all offspring will physically appear tall and yellow-seeded.

5. **Correct Answer: B**

In human reproduction, there is always a 1:1 ratio of X-carrying sperm to Y-carrying sperm; therefore, every single fertilization event has an exactly 50% (1/2) statistical probability of producing a male child.

8. OUR ENVIRONMENT

Solutions

Worksheet – 1

1. **Correct Answer: B) Structural and functional**
   Explanation: An ecosystem is the structural and functional unit of nature, consisting of interactions between biotic (living) and abiotic (non-living) components.
2. **Correct Answer: B) A. G. Tansley**
   Explanation: The term "ecosystem" was coined by A. G. Tansley in 1935.
3. **Correct Answer: B) Producers and consumers**
   Explanation: The basic trophic levels are producers (autotrophs) and consumers (heterotrophs); decomposers recycle nutrients but are not a primary trophic level.
4. **Correct Answer: B) A man-made ecosystem created by humans**
   Explanation: Anthropogenic ecosystems are created and maintained by humans, such as agricultural fields, gardens, and aquaria.
5. **Correct Answer: B) A step in a food chain**
   Explanation: A trophic level represents each step in a food chain, indicating the position of organisms based on their feeding relationships.
6. **Correct Answer: B) Grasshopper**
   Explanation: Grasshoppers are herbivores that feed on plants and are therefore primary consumers.
7. **Correct Answer: C) Feed directly on producers**
   Explanation: Primary consumers obtain energy by feeding directly on producers (plants or autotrophs).
8. **Correct Answer: C) Sun**
   Explanation: The Sun is the ultimate source of energy for almost all ecosystems, as it drives photosynthesis in producers.
9. **Correct Answer: D) Aquarium**
   Explanation: An aquarium is a human-made ecosystem and therefore an example of an anthropogenic ecosystem.
10. **Correct Answer: B) Primary consumers**
    Explanation: Secondary consumers are carnivores or omnivores that feed on primary consumers (herbivores).

Worksheet-2

1. **Correct Answer: C) Grazing, detritus and parasitic**

   Explanation: Food chains are classified based on how energy flows: grazing (plants → herbivores → carnivores), detritus (dead organic matter → decomposers), and parasitic (host → parasite).
2. **Correct Answer: C) Begins with dead organic matter and ends with decomposers**

   Explanation: Detritus food chain starts with dead organic matter and passes energy to decomposers and detritivores.
3. **Correct Answer: C) Food web**

   Explanation: A food web is a network of interlinked food chains showing multiple feeding relationships at different trophic levels.
4. **Correct Answer: C) Begins with a producer and ends with a consumer**

   Explanation: Grazing food chain starts with producers (plants) and passes energy to herbivores and then to carnivores.
5. **Correct Answer: C) About 10% of energy is transferred from one trophic level to the next**

   Explanation: According to Lindeman's 10% law, only about 10% of energy is transferred from one trophic level to the next; the rest is lost as heat or used in metabolism.

6. **Correct Answer: C) Parasitic food chain**
Explanation: In a parasitic food chain, energy flows from a host organism to a parasite.
7. **Correct Answer: C) Biomagnification**
Explanation: Biomagnification is the increase in concentration of non-degradable chemicals (like pesticides) at higher trophic levels.
8. **Correct Answer: B) Predator–prey relationships and population size**
Explanation: Food webs help regulate populations and maintain balance between predators and prey.
9. **Correct Answer: A) Growth, balance, and overall development of the ecosystem**
Explanation: Food webs contribute to the overall stability, nutrient cycling, and balance of an ecosystem.
10. **Correct Answer: C) 1%**
Explanation: In terrestrial ecosystems, plants capture about 1% of the sunlight energy for photosynthesis; the rest is reflected or lost.

Worksheet-3

1. **Correct Answer: B) Eltonian pyramid**
Explanation: Ecological pyramids are also called Eltonian pyramids, named after Charles Elton, showing energy flow, numbers, or biomass in trophic levels.
2. **Correct Answer: B) Excessive deposition of organic matter and nutrients**
Explanation: Eutrophication occurs due to excessive nutrients (like nitrogen and phosphorus) in water, leading to algal blooms and oxygen depletion.
3. **Correct Answer: B) Use of pesticides and other chemicals**
Explanation: Biomagnification occurs when non-biodegradable chemicals (e.g., DDT, pesticides) accumulate and increase in concentration at higher trophic levels.
4. **Correct Answer: C) Pyramid of energy**
Explanation: The pyramid of energy is always upright because energy decreases at higher trophic levels due to loss as heat and metabolism.
5. **Correct Answer: B) Absorbs the sun's harmful UV radiation**
Explanation: The ozone layer in the stratosphere protects life by absorbing most of the sun's harmful ultraviolet (UV) radiation.
6. **Correct Answer: B) Chlorofluorocarbons (CFCs)**
Explanation: CFCs released from aerosols and refrigerants are the primary cause of ozone depletion.
7. **Correct Answer: B) United Nations Environment Programme**
Explanation: UNEP stands for United Nations Environment Programme, which coordinates global environmental activities.
8. **Correct Answer: C) Montreal Protocol**
Explanation: The Montreal Protocol (1987) is an international agreement to phase out ozone-depleting substances like CFCs.
9. **Correct Answer: B) Water bodies**
Explanation: Eutrophication affects water bodies by increasing nutrient load, causing algal blooms and oxygen depletion.
10. **Correct Answer: C) Water pollution**
Explanation: Eutrophication is a form of water pollution caused by excessive nutrients in aquatic ecosystems.

Competitive Corner

SINGLE CORRECT OPTION TYPE

1. **Correct Answer: C) Interaction between biotic and abiotic components**
Explanation: An ecosystem consists of living organisms (biotic) and their physical environment (abiotic) interacting as a functional unit.
2. **Correct Answer: B) Develops without human interference**
Explanation: A natural ecosystem forms on its own through ecological processes, without human intervention.
3. **Correct Answer: D) Producer**
Explanation: Food chains begin with producers (plants or autotrophs) that synthesize their own food from sunlight.

4. **Correct Answer: C) Grass → rabbit → hawk**
   Explanation: A grazing food chain starts with producers, followed by herbivores, and then carnivores.
5. **Correct Answer: D) Energy, biomass, or number of individuals**
   Explanation: Ecological pyramids can represent energy flow, biomass, or the number of organisms at each trophic level.
6. **Correct Answer: B) Organic matter**
   Explanation: Eutrophication is caused by excessive deposition of organic matter and nutrients (like nitrogen and phosphorus) in water bodies.
7. **Correct Answer: B) They occupy different niches on the same tree**
   Explanation: Each bird species exploits a different part of the tree and different food sources, reducing direct competition. This is an example of niche differentiation.
8. **Correct Answer: C) Small fish**
   Explanation: Small fish feed directly on algae, so a decline in algae affects them first and most directly. Large fish are affected later because their food (small fish) is impacted.
9. **Correct Answer: B) Different feeding heights and food types**
   Explanation: Rabbits feed on grass near the ground while goats browse on shrubs. This reduces competition, allowing coexistence due to resource partitioning.
10. **Correct Answer: B) Reduced plant growth due to niche overlap**
    Explanation: The new insect species and aphids both consume plant sap. Initially, the overlap in resource use can stress the plants and reduce growth.

ONE OR MORE THAN ONE CORRECT OPTION TYPE

11. **Correct Answers: A), B), D)**
    Explanation: An ecosystem is the structural and functional unit of nature, it is self-regulating and self-sustaining, and it involves reciprocal interactions between biotic and abiotic components. It does not include only living organisms.
12. **Correct Answers: A), B)**
    Explanation: Anthropogenic ecosystems are created or modified by humans, such as agriculture fields and aquariums. Natural ecosystems like marine or forest ecosystems are not anthropogenic.
13. **Correct Answers: A), B), D)**
    Explanation: The pyramid of energy represents energy flow at each trophic level, is always upright, and shows energy decrease at higher trophic levels. It cannot be inverted, even in aquatic ecosystems.
14. **Correct Answers: A), B), D)**
    Explanation: A parasitic food chain starts with a host, ends with a parasite, and involves living organisms. It does not start with dead organic matter.
15. **Correct Answers: A), C), D)**
    Explanation: CFCs contain carbon, chlorine, and fluorine, are used in refrigerators, and contribute to ozone layer depletion by preventing ozone from reforming. They do not help in ozone formation.

ASSERTION AND REASON TYPE

16. **Correct Answer: C**
    Explanation: The assertion is correct — a food web is a network of interconnected food chains. The reason is incorrect — a kite can indeed be part of a food web as a predator.
17. **Correct Answer: D**
    Explanation: Trophic levels include all organisms (producers, consumers, decomposers), not just plants. The reason is correct.
18. **Correct Answer: D**
    Explanation: Producers convert inorganic compounds into organic compounds (via photosynthesis), not the other way. The reason about solar energy is correct but does not justify the incorrect assertion.
19. **Correct Answer: A**
    Explanation: Eutrophication leads to algal blooms and fish kills, and it is caused by excessive deposition of dissolved and particulate organic matter — the reason correctly explains the assertion.

MATRIX MATCHING TYPE

20. **Correct Answer: A**
21. **Correct Answer: A**

COMPREHENSION TYPE

Passage-1

22. **Correct Answer: C – Stratosphere**
23. **Correct Answer: B- Absorb harmful UV radiation**
24. **Correct Answer: C Three oxygen atoms**

Passage -2

25. **Correct Answer: B – Energy**
26. **Correct Answer: C – Trophic level**
27. **Correct Answer: C – Detritus food chain**

PREVIOUS YEAR QUESTIONS

1. **Correct Answer: D**
   Kyoto Protocol (Correct year: 1997, not 1982)
2. **Correct Answer: D**
   (i), (ii), (iii), and (iv) — all can result from ozone depletion
3. **Correct Answer: C**
   Tissue culture (not a conservation method in natural habitats)
4. **Correct Answer: D**
   Oxygen is not a greenhouse gas
5. **Correct Answer: D**
   Jute bag, mouldy bread, newspaper (all biodegradable)
6. **Correct Answer: A**
   Undirectional energy flow (represents energy transfer in a food chain)
7. **Correct Answer: C**
   Pyramid of numbers is upright in grassland, inverted in parasitic food chain, not the other way around
8. **Correct Answer: D**
   Aerosol cans do not add methane; they release CFCs
9. **Correct Answer: B**
   Both statements are true
10. **Correct Answer: A**
    Bhakra Dam – Satluj river